On The Gravitational Nature Of Time

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Abstract

We explore a modified set of coordinate transformations in which time, cosmic inflation and gravitational acceleration are treated as the same process observed from different reference frames, using an unwrapped Kaluza-Klein-like geometry with a fourth spatial coordinate. The transformations preserve the Minkowski interval for instantaneous events and retain a classical notion of simultaneity by assigning the additional spatial separation to this coordinate. Within this framework, a trigonometric construction yields an orbital velocity equation that, using only the enclosed baryonic mass of the Milky Way, matches the observed solar-neighborhood circular velocity to within **5.6%**. The construction also yields two characteristic velocities, a kinematic velocity and a cosmological velocity, which we compare with CMB dipole and supernova-based measurements as a possible geometric account of the discrepancy between them. We further show that the induced radial metric can reproduce the spatial geometry of the Schwarzschild solution. The model remains incomplete: it has not been unified with general relativity, a temporal metric relation is still required, and the dimensional treatment of the gravitational relations needs further formalization. We propose the framework as an alternative model of chronology that may avoid selective reference-frame preference, and outline directions for further work.

Keywords: gravitation, time, relativity, inflation

1 Introduction

Albert Einstein was insistent on a static Universe during the period in which he wrote his specialized theory of relativity[1–4](SR). What is proposed here is largely a re-examination of the geometry of SR under the assumption that the $\mathbb{R}^3$ space between two arbitrary points elongates as time progresses.

This dilation is then bound to γ, giving a velocity dependent function for $\vec{g}$ and a magnitude that extends into a 4th spatial coordinate, y, similar in construction to an unwrapped Kaluza-Klein (KK) dimension[5, 6]. Through these coordinate transformations time, cosmic inflation and gravitational acceleration become the same process as observed from different reference frames in a space-time geometry where the 4th spatial KK dimension acts as a concurrent density axis.

This geometry gives an alternative model of concurrency and synchronicity as a classical notion of simultaneity is maintained by attributing the additional spatial separation to the 4th spatial dimension, $dy = v\tau\sqrt{\gamma^2 - 1}$ as shown in figure 1. While maintaining classical simultaneity resolves numerous temporal paradoxes without the need for selective reference frame preference, it produces a concurrent axis that describes the ultra-dense state of the early Universe as being separated only by some Δ along this 4th spatial axis.

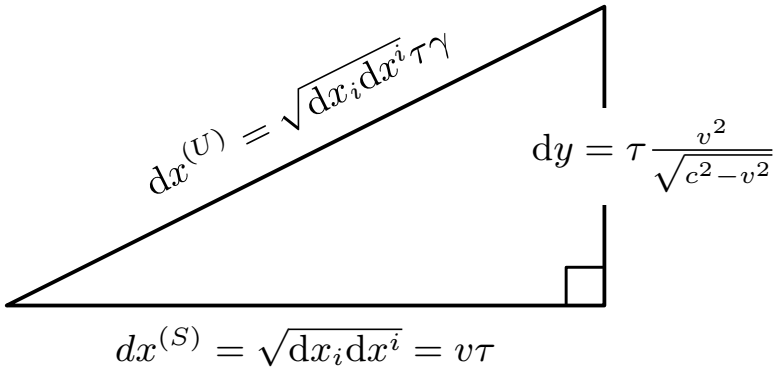


Fig. 1 Right-triangle construction used to represent the ordinary spatial displacement dx, the additional spatial displacement dy, and the extended displacement $dx^{(U)}$. The relation assigned to the hypotenuse is an assumption of the proposed geometric construction.

1.1 The Lighthouse & The Clock-Tower

Let an event be defined classically, such that a single, instantaneous event represents a local measurement as $dt \to 0$.

Let there be two overlapping $\mathbb{R}^3$ coordinate systems, **S** and **U**. Let **S** and **U** represent an equi-temporal surface within a 4-dimensional hyperspace where $y = y^{(S)}$ and $y = y^{(U)}$ in each respective 3-dimensional surface.

This gives

$$f\left(x_i, y\right) = \partial_y \mathcal{H} \tag{1}$$

where

$$f\left(x_i, y^{(A)}\right) = \mathbf{A}\left(x_i\right) \tag{2}$$

for an arbitrary $\mathbf{A}$ contained within $\mathcal{H}$.

Let $\mathbf{U}$ represent a 'future' state from the perspective of an observer in $\mathbf{S}$ so that

$$v^{(S)} = (t, x_\mu) \qquad v^{(U)} = (t + \Delta t, x_\mu) \tag{3}$$

where $\Delta t > 0$ and $\mu = 1, 2, 3, 4$. Likewise, let $\mathbf{S}$ and $\mathbf{U}$ be bound such that

$$f\left(x_i, y^{(U)}\right) = f\left(x_i, y^{(S)}\right) + \int_{x_0^{(S)}}^{x_0^{(U)}} F\left(x\right)\ dt \tag{4}$$

where F is the sum of all forces acting on the point of space at x, integrated according to time, which in turn sums the change between the temporal surfaces $\mathbf{S}$ and $\mathbf{U}$.

Let there also be two observers, A and B. Let A and B agree that B should travel at some *agreed upon* relativistic velocity v' between two arbitrary points in space contained within $\mathcal{H}$, p_1 and p_2. Let the experiment be arranged such that B passes p_1 in $\mathbf{S}$, and p_2 in $\mathbf{U}$.

Let A remain at relative rest with respect to p_1, and at p_2 place a time keeping device large enough to be visible from p_1.

We then make these assertions

1. To maintain a classical model of simultaneity A and B must agree upon the time at which B reaches p_2.
2. To satisfy existing experimental evidence for the additional γ factor[7–11], $\Delta t = \gamma \frac{d}{v}$.

This gives a set of coordinate transformations such that for each relative observer

$$\Delta t = \frac{\overbrace{d \cdot \gamma}^{B}}{v^{(U)}} = \underbrace{\frac{d}{v^{(S)}} = \frac{d}{v^{(U)}\left(\frac{1}{\gamma}\right)}}_{A} \tag{5}$$

2 Theoretical Framework

We acknowledge two primary properties of time:

1. A driving rate of change, at least as fundamental as the $\mathbb{R}^3$ dimensions.
2. A 4th displacement coordinate.

Throughout the following derivations, we presume that time has no physical significance beyond the two properties described above. We presume that time acts only as a mathematical tool: $\Delta t = \frac{d}{v}$. In turn, we must attribute the 4th displacement coordinate to a position along the density axis that arises, y, and the driving rate of change to *motion*.

Likewise, we presume that in a co-moving frame of reference along the y axis, mechanical equations should limit at their Newtonian counterpart with only the relativistic portion of associated magnitudes protruding out of the containing temporal surface.

To remove time as a physically meaningful quantity, we let

$$\tau = \frac{\dot{x}}{\widehat{\dot{x}}} \tag{6}$$

where $\dot{x}$ indicates not a derivative with respect to time, but rather a fundamental driving rate of change, a magnitude of length that grows with an arbitrary temporal scalar. Let $\widehat{\dot{x}} = \partial_\tau x \ \hat{\tau} = |\vec{x}|$ where $\hat{\tau} = 1$, or the velocity in units of distance. τ, while equivalent in magnitude to t is derived directly from the geometry and has no explicit time dependence.

2.1 Lorentz Invariance

We consider these coordinate transformations with regards to the Lorentz transformations and find that for an instantaneous event where $dt \to 0$

$$\boxed{\begin{aligned} dx' &= \gamma dx = v^{(U)} \\ c\,dt' &= -\gamma \frac{v}{c} dx = -\partial_\tau y \end{aligned}} \tag{7}$$

where y is as defined in equation 9.

We define $v\,(\gamma)$ to be the Lorentz factor solved for v

$$v\,(\gamma) = c\sqrt{1 - \frac{1}{\gamma^2}} \tag{8}$$

Beginning with the definition of the additional coordinate,

$$y = \tau \frac{v^2}{\sqrt{c^2 - v^2}} \tag{9}$$

we take the partial derivative with respect to τ, holding v constant:

$$\boxed{\partial_\tau y = \frac{v^2}{\sqrt{c^2 - v^2}}} \tag{10}$$

Since $x = v\tau$, we can equivalently express the coordinate as a function of x:

$$y = x \frac{v}{\sqrt{c^2 - v^2}}. \tag{11}$$

Taking the partial derivative with respect to x gives

$$\boxed{\partial_x y = \frac{v}{\sqrt{c^2 - v^2}}} \tag{12}$$

We can now express this derivative in terms of the Lorentz factor:

$$\partial_x y = \frac{\gamma v}{c}. \tag{13}$$

Multiplying by the velocity expressed as a function of γ,

$$\begin{aligned} v(\gamma)\partial_x y &= c\sqrt{1-\frac{1}{\gamma^2}}\frac{\gamma v}{c} \\ &= v\sqrt{\gamma^2-1} \\ &= \frac{\gamma v^2}{c}. \end{aligned} \tag{14}$$

Finally we recover

$$\boxed{v(\gamma)\partial_x y = \frac{\gamma v^2}{c} = \partial_\tau y} \tag{15}$$

2.1.1 Invariance of the Minkowski Interval

We can further examine the Lorentz invariance of these transformations by considering the Minkowski interval. For an instantaneous event, where $dt \to 0$, the transformed interval is given by

$$ds'^2 = c^2 dt'^2 - dx'^2. \tag{16}$$

Substituting the coordinate transformations established above,

$$\begin{aligned} ds'^2 &= \left(-\gamma\frac{v}{c}dx\right)^2 - (\gamma dx)^2 \\ &= \gamma^2 dx^2\left(\frac{v^2}{c^2}-1\right). \end{aligned} \tag{17}$$

Using the definition of the Lorentz factor,

$$\gamma^2 = \frac{1}{1-\frac{v^2}{c^2}}, \tag{18}$$

we obtain

$$ds'^2 = -\gamma^2 dx^2\left(1-\frac{v^2}{c^2}\right). \tag{19}$$

Therefore,

$$\boxed{ds'^2 = -dx^2 = ds^2} \tag{20}$$

Thus, the instantaneous coordinate transformations preserve the Minkowski interval, demonstrating consistency with the invariance condition of Lorentz transformations.

2.1.2 The Lorentz Factor as a Coordinate Gradient

Having established the relationship between the additional coordinate and the Lorentz transformations, we can rearrange the spatial derivative to express the Lorentz factor directly in terms of $\partial_x y$.

As shown in equation 65,

$$\partial_x y = \frac{\gamma v}{c} = \beta\gamma \tag{21}$$

Squaring both sides gives

$$(\partial_x y)^2 = \frac{\gamma^2 v^2}{c^2} \tag{22}$$

Using the definition of the Lorentz factor,

$$\gamma^2 = \frac{1}{1 - \frac{v^2}{c^2}}, \tag{23}$$

we can rewrite this expression as

$$(\partial_x y)^2 = \frac{v^2}{c^2 - v^2}. \tag{24}$$

Rearranging and substituting back into the Lorentz factor gives

$$\begin{aligned} \gamma^2 &= 1 + \frac{v^2}{c^2 - v^2} \\ &= 1 + (\partial_x y)^2. \end{aligned} \tag{25}$$

Consequently, we obtain the following expression for the Lorentz factor in terms of the additional coordinate:

$$\boxed{\gamma = \sqrt{1 + (\partial_x y)^2}} \tag{26}$$

This establishes a direct relationship between the Lorentz factor and the spatial gradient of the additional coordinate, allowing the relativistic factor to be expressed geometrically without explicitly introducing velocity.

2.2 The Equivalence Principle

Here we examine these coordinate transformations in the context of the equivalence principle. While a well defined function remains elusive, these generialized coordinate transformations and a numerical intersection with the fine structure constant allows us to derive several important properties of an apparent electromagnetic function producing $\vec{g}$.

2.2.1 Earth's Kinematic Velocity From Electromagnetic Parameters

We define $\delta(M, R)$ to be

$$\delta\left(M,R\right)=\frac{d}{dR}g=2\frac{GM}{R^3} \tag{27}$$

If $\delta\left(M,R\right)$ is induced by the coordinate transformations described above, we should presume that this dilation is radial. We then find the magnitude of a single $\mathbb{R}^3$ axis by taking the inverse of the determinant of a diagonal 3×3 matrix:

$$\omega\left(M,R\right)=\sqrt[3]{\frac{d}{dR}g}=\left(\delta\left(M,R\right)\right)^{1/3} \tag{28}$$

Investigating this quantity for Earth gives

$$\omega_{\oplus}=0.014537\ldots \tag{29}$$

While this approximates 2α, it more closely produces

$$\omega_{\oplus}=\alpha\left(2-\alpha\right)\pm 0.0253\% \tag{30}$$

Then by expanding equation 30

$$\alpha\left(2-\alpha\right)=\sqrt[3]{\frac{d}{dR}g} \tag{31}$$

Moving the electromagnetic parameters and R to one side before integrating with respect to R gives

$$R^4\left(\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3}+\frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5}-\frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4}-\frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6}\right)=1.272\times 10^{21} \tag{32}$$

We then take the radius independent magnitude, isolating the additional electromagnetic parameters

$$\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3}+\frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5}-\frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4}-\frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{33}$$

As γ represents 1 plus some small ds, let us add 1 to these additional parameters and substitute γ_{em} into equation 8.

$$\gamma_{\mathrm{em}}=1+\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3}+\frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5}-\frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4}-\frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{34}$$

$$v_{(\gamma_{\mathrm{em}})}=371,721\ \mathrm{m/s} \tag{35}$$

This matches the 2018 Planck data[12] to within 0.514%, and is in rough agreement with general observation of the velocity necessary to induce the CMB dipole.

2.2.2 Solar ω

We expand equation 31 and define $\Gamma(\sigma)$ to produce

$$\alpha = \Gamma(\sigma)\left(\frac{d}{dR}g\right)^{1/3} \tag{36}$$

Where $\Gamma(\sigma)$ is some unknown function of spin. While a well defined function remains elusive, we can take advantage of several characteristics for specific local bodies according to table 1 to derive certain characteristics of σ.

Table 1 Advantageous properties

Body	Trait	Value
Earth	The most mass dense.	5495 kg m^{-3}
Saturn	The least mass dense.	687 kg m^{-3}
Sun	The most massive.	1.988×10^{30} kg

In equation 31 we've demonstrated that for the most mass dense body in the system

$$\Gamma_{\oplus}(\sigma) \approx \Gamma(\alpha) = \frac{1}{2-\alpha} \tag{37}$$

We propose that $\Gamma(\sigma)$ acts inside of the determinant. We then take the inverse of the determinant of a diagonal 3×3 matrix

$$\Gamma_{\odot}(\sigma) \propto \sqrt[3]{\Gamma(\sigma)} \tag{38}$$

We then presume that σ is associated with spin. Due to the increased surface gravitational acceleration and the decreased rate of spin for the Sun, we approximate $\sigma = 0$.

$$\Gamma_{\odot}(\sigma) \approx \frac{1}{\sqrt[3]{2}} \tag{39}$$

Or

$$\alpha = \Gamma_{\odot}(\sigma)\sqrt[3]{\frac{d}{dR}g} = \sqrt[3]{\frac{1}{2}\frac{d}{dR}g_{\odot}} \pm 0.47\% \tag{40}$$

We then find through observation that for the least mass dense body in the system, $\Gamma_{\hbar}(\sigma) \approx 1$.

This gives a set of 3 equations where the $\Gamma(\sigma)$ term cancels for the dominant body in the system.

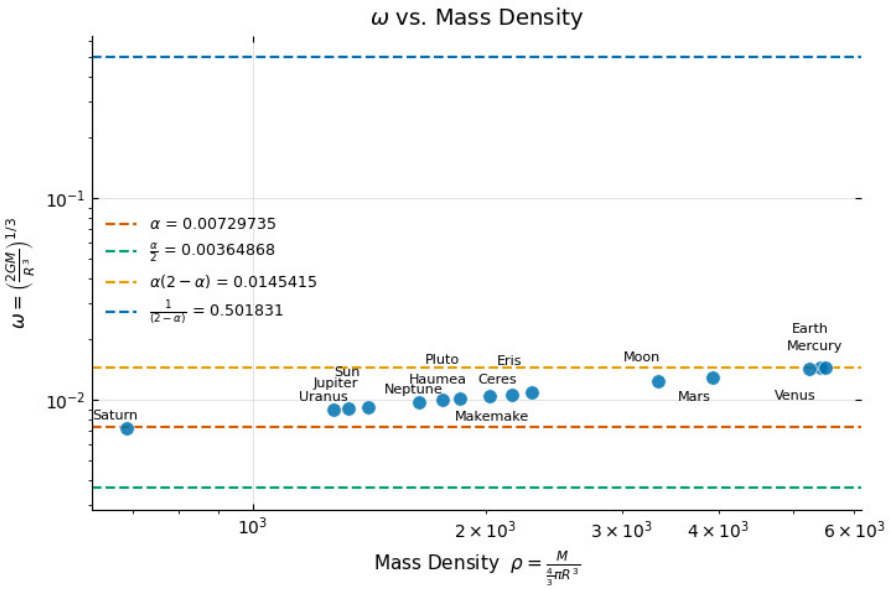


Fig. 2 The model quantity ω, identified with dy and constructed from the inverse determinant associated with volume change, plotted as a function of mass density.

$$\begin{aligned} g_{\oplus} &= R_{\oplus}\left(\frac{\alpha^3\,(2-\alpha)^3}{2}\right)\tau \quad && \pm 0.008\% \\ g_{\odot} &= R_{\odot}\alpha^3\tau && \pm 1.426\% \\ g_{\hbar} &= \frac{1}{2}R_{\hbar}\alpha^3\tau && \pm 1.134\% \end{aligned} \tag{41}$$

Which in turn produces a domain where $\alpha \lessapprox \omega \lessapprox \alpha\,(2-\alpha)$ as shown in figure 2.

2.3 Gravitational Parameters

Due to algebraic symmetries, it's trivial to derive our kinematic velocity through gravitational parameters. What is of interest however, is that we derive what we propose is a more fundamental velocity first. It is only after integrating motion in a manner that is analogous to time that we produce our kinematic velocity. We call this first velocity our *cosmological velocity*, and propose a new geometric interpretation of existing data that may resolve the SNe/CMB velocity discrepancy.

2.3.1 Cosmological Velocity

As γ is applied as a scalar, we take the average of the integral of $\delta_{(M,R)}$ to be

$$\varphi\,(M,R) = \frac{1}{R}\int \delta\,(M,R) = 1 + G\frac{M}{R^3} \tag{42}$$

Substituting $\varphi\,(M, R)$ into equation 8 gives

$$\boxed{v_{\text{cosmo}} = c\sqrt{1 - \frac{1}{\varphi^2(M,R)}} = 525,493.17 \text{ m/s}} \tag{43}$$

While this value aligns closely with the correlation carried out by Gordon et al.[13], there remains large uncertainties inherent in SNe observation[14, 15].

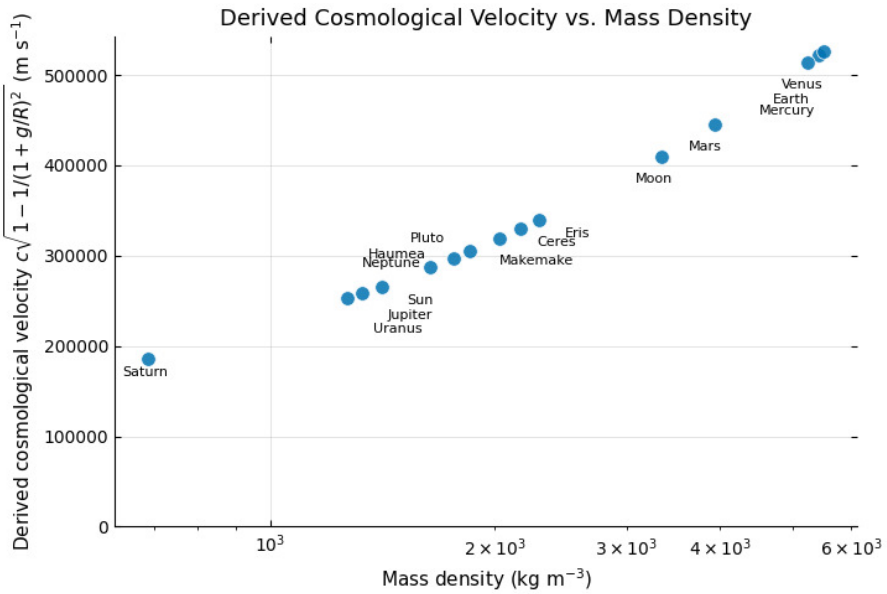


Fig. 3 Model-derived cosmological velocity plotted as a function of mass density.

As shown in figure 3, this function only holds for the theoretical upper bounds of mass density. We speculate that there may exist some yet unknown process in the formation of solar systems, at least at our scale and location within the parent galaxy that leads to a natural function of mass density and velocity. While the details of this process have yet to be resolved, the coordinate transformations described here intrinsically link mass density to velocity as $\vec{g}$ becomes a function of v.

2.3.2 Kinematic Velocity

We apply the time-equivalent quantity in equation 6 to equation 42, making equation 42 time dependent. Integrating gives

$$\frac{1}{R}\int_0^R \int_0^x \delta\tau \; dR \; d\tau = 1 + \frac{1}{2}\frac{GM_\oplus}{R_\oplus^3}\left(\frac{\dot{x}}{\hat{x}}\right)^2 \tag{44}$$

We then substitute equation 44 into equation 8 where $d\tau = 1$

$$\boxed{v_{\text{kinematic}} = c\sqrt{1 - \frac{1}{\left(1 + \frac{1}{2}G\frac{M}{R^3}\right)^2}} = 371,580 \text{ m/s}} \tag{45}$$

which matches the 2018 Plank data to within 0.47%.

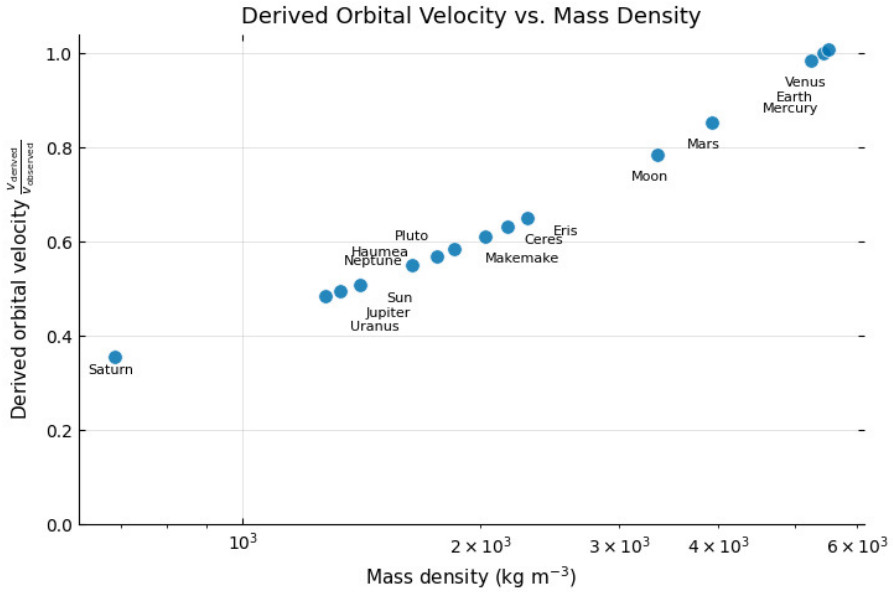


Fig. 4 Model-derived kinematic velocity as a ratio with the velocity found by Eilers et al. plotted as a function of mass density.

2.4 Orbital Velocity

2.4.1 Relativistic Stable Orbit Velocity

In the simplified single-body system considered here, the spatial dilation described above permits a trigonometric derivation of the stable-orbit velocity as an alternative to the conventional algebraic equality.

Figure 5 represents the geometric construction used in this derivation.

$$R_0^2 + v_{\text{orbit}}^2 d\tau^2 = \left(\int_0^{R_0} 2G\frac{M}{R^3}\right)^2 = (R_0 + g\ d\tau)^2 \tag{46}$$

Setting $\tau = 1$ gives

$$\boxed{v_{\text{orbit}}\,(M, R) = \frac{\sqrt{GM\,(GM + 2R^3)}}{R^2}} \tag{47}$$

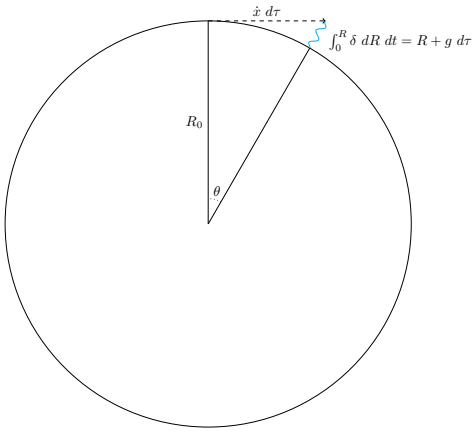


Fig. 5 Geometric construction used in the proposed stable-orbit derivation, showing R_0, the angle θ, the displacement $\dot{x}\,d\tau$, and the proposed radial increment $R + g\,d\tau$.

For the comparison with the Milky Way, we adopt the enclosed Galactic baryonic mass reported by Cautun et al. [16]:

$$M_{\text{enc}} = M_{\text{bar}}(< R_0) = 4.44 \times 10^{10}\, M_{\odot}, \tag{48}$$

Here, $R_0 = 8.12\,\text{kpc}$ denotes the Sun's Galactocentric orbital radius[17, 18].

For comparison with observation, we use the solar-neighborhood circular velocity reported by Eilers et al. [19]. Using the adopted enclosed mass and solar orbital radius, the model gives the following orbital velocity:

$$v_{\text{orbit}}\,(M_{\text{enc}}, R_{\text{orbit}}) = 216,874\ \text{m/s} = v_{\text{eilers}} \pm 5.59\% \tag{49}$$

The model velocity equals the comparison velocity at $M_{\text{enc}} \approx 4.95 \times 10^{10} M_{\odot}$.

The adopted mass scale is also consistent with the characteristic stellar mass scale of approximately $5 \times 10^{10}\, M_{\odot}$ identified by Yu et al. [20] in their FIRE-2 simulations of Milky Way-mass galaxies. Yu et al. associate this scale with a transition in galactic morphology from predominantly disc-dominated to increasingly spheroid-dominated systems.

It is important to note here that $v\,(M, R) \approx v_{\text{escape}}$ at $R = R_{\oplus}$ and $M = M_{\oplus}$, not a stable orbit velocity. This discrepency needs to be accounted for by Machian resistance in an n-body system. The associated equations remain elusive, but geometric unity achieved elsewhere appears to indicate the viability of this geometry, such that the sum of this radiating force for all other bodies in the system may induce inertial resistance.

2.4.2 Kinematic Velocity From Triginometric Orbit Equations

We now investigate equation 49 by setting

$$v_{\text{orbit}} = \sqrt{2\frac{GM}{R}} \tag{50}$$

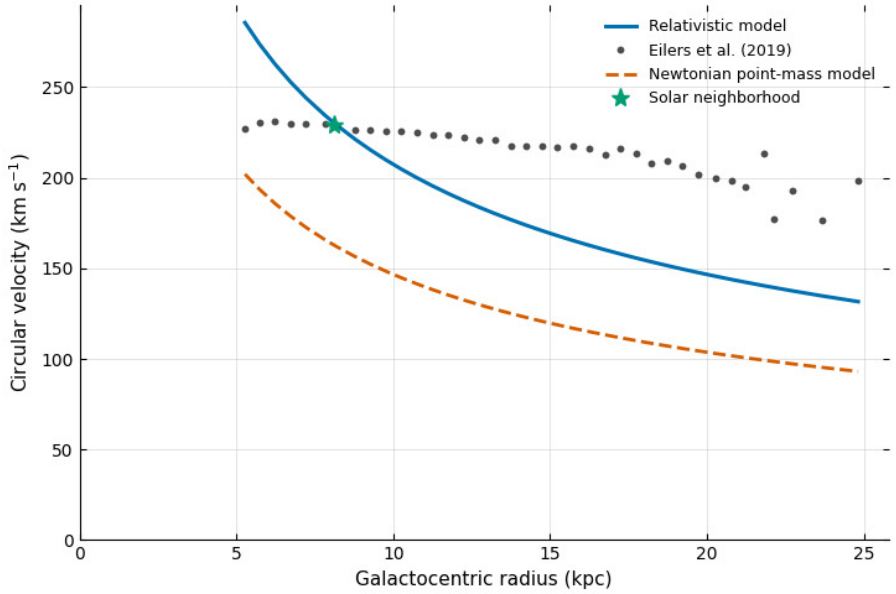


Fig. 6 Milky Way circular velocity as a function of galactocentric radius. Eilers et al. (2019) measurements are shown as points, alongside the relativistic and Newtonian point-mass model curves. The solar-neighborhood measurement is marked with a star; error bars indicate the reported uncertainties in galactocentric radius and circular velocity.

which is obtained by $E_{\text{kinetic}} = \int g \, dR$. Solving equation 46 for τ and substituting v_{orbit} gives

$$dt = \frac{2R^3}{GM - 2R^3} = -1.000000768 \tag{51}$$

Inserting this quantity into equation 8 gives

$$v_{(|dt|)} = 371,580 \text{ m/s} \tag{52}$$

which matches the 2018 Plank data to within 0.47%.

2.5 Fifth-Dimensional Spatial Displacement and Gravitational Force

While most of the derivations to this point have relied on simplified single body solutions, we explore the connections to the KK 5-dimensional metric here. As the KK model imposes a cylindrical condition largely due to the lack of an observable 4th spatial dimension, we derive the magnitude of this 4th dimension from the coordinate transformations described above and apply no such condition.

2.5.1 Geometric Construction of the Fifth-Dimensional Displacement

We extend the ordinary three-dimensional displacement into the additional spatial coordinate y. Let dx denote the magnitude of the ordinary spatial displacement,

$$dx = \sqrt{dx_i dx^i}, \qquad i = 1, 2, 3. \tag{53}$$

For spatial velocity magnitude v and displacement parameter τ, as defined in the theoretical framework, the ordinary displacement is

$$dx = v\tau. \tag{54}$$

The proposed construction treats the extended displacement $dx^{(U)}$ as the hypotenuse of a right triangle whose perpendicular components are dx and dy as shown in figure 1. With γ denoting the Lorentz factor defined in Equation 8,

$$dx^{(U)} = v\tau\gamma. \tag{55}$$

Applying the Pythagorean theorem,

$$\left(dx^{(U)}\right)^2 = dx^2 + dy^2, \tag{56}$$

substitution gives

$$dy^2 = (v\tau\gamma)^2 - (v\tau)^2 \tag{57}$$

$$= v^2\tau^2\left(\gamma^2 - 1\right). \tag{58}$$

Taking the positive magnitude of the additional displacement yields

$$dy = v\tau\sqrt{\gamma^2 - 1} = \tau\frac{v^2}{\sqrt{c^2 - v^2}}. \tag{59}$$

The velocity form follows from the Lorentz-factor relation in Equation 8.

For constant v, differentiation with respect to τ gives the rate of displacement along y:

$$\boxed{\partial_\tau y = \frac{v^2}{\sqrt{c^2 - v^2}}} \tag{60}$$

2.5.2 Incorporation into the Kaluza–Klein Metric

We represent the additional spatial coordinate through the metric component G_{44}. For a diagonal five-dimensional metric, the line element is

$$dS^2 = g_{\mu\nu}dx^\mu dx^\nu + G_{44}dy^2. \tag{61}$$

The corresponding metric tensor has the block structure

$$\boxed{G_{AB} = \begin{pmatrix} g_{\mu\nu} & 0 \\ 0 & G_{44} \end{pmatrix}} \tag{62}$$

The component G_{44} sets the metric scaling of the additional coordinate. Its radial variation determines the coupling between motion along y and radial motion in the ordinary spatial coordinates.

For radial coordinate R, the Christoffel symbol associated with this coupling is

$$\Gamma^R_{44} = -\frac{1}{2} g^{RR} \frac{\partial G_{44}}{\partial R}. \tag{63}$$

For a normalized radial coordinate, $g^{RR} = 1$, so

$$\boxed{\Gamma^R_{44} = -\frac{1}{2} \frac{\partial G_{44}}{\partial R}} \tag{64}$$

2.5.3 Finding G_{44}

We take

$$\partial_x y = \frac{v}{\sqrt{c^2 - v^2}} \tag{65}$$

Or

$$dy = \tau \frac{v}{c^2 - v^2} dx \tag{66}$$

We then take the KK line element to be

$$dS^2 = g_{\mu\nu} dx^\mu dx^\nu + G_{44} dy^2 \tag{67}$$

and the $\mathbb{R}^4$ spatial separation to be

$$dS^2_{\text{spatial}} = dx^2 + dy^2 \tag{68}$$

where $dx^2 = dx_i dx^i$ for $i = 1, 2, 3$. Expanding this gives

$$dS^2_{\text{spatial}} = dx^2 \left(1 + \frac{v^2}{c^2 - v^2}\right) = dx^2 \gamma^2 \tag{69}$$

which matches the expected displacement magnitude of the Lorentz transformations.

Applying a flat metric to the line element defined in equation 67 gives

$$dS^2_U = dx^2 + G_{44} dy^2 \tag{70}$$

yielding

$$\boxed{G^{(U)}_{44} = 1} \tag{71}$$

for $dS^2_U = \gamma^2 dx^2$.

However solving equation 70 for G_{44} in the **S** coordinate system where $dS_S^2 = d_i d^i$ gives

$$\boxed{G_{44}^{(S)} dy^2 = 0} \tag{72}$$

indicating that Newtonian mechanics may remain contained within a temporal surface, while the relativistic portion of displacement magnitude extends into the 4th spatial dimension, y. For this reason, we speculate that inertia has it's origins purely in the relativistic component of displacement; a consequence of a Machian-like resistance along the 4th spatial dimension.

2.6 The Four-Potential

2.6.1 Restoring the Off-Diagonal Terms

The block-diagonal form of G_{AB} above sets $G_{\mu 4} = 0$. In Kaluza–Klein geometry the off-diagonal components are the four-potential. We restore them by writing the line element as

$$dS^2 = g_{\mu\nu}\, dx^\mu dx^\nu + G_{44}\,(dy + \mathcal{A}_\mu dx^\mu)^2 \tag{73}$$

so that

$$G_{AB} = \begin{pmatrix} g_{\mu\nu} + G_{44}\mathcal{A}_\mu\mathcal{A}_\nu & G_{44}\mathcal{A}_\mu \\ G_{44}\mathcal{A}_\nu & G_{44} \end{pmatrix}, \qquad G^{\mu\nu} = g^{\mu\nu}, \quad G^{\mu 4} = -\mathcal{A}^\mu, \quad G^{44} = \frac{1}{G_{44}} + \mathcal{A}_\mu\mathcal{A}^\mu. \tag{74}$$

The line element is invariant under $y \to y + \Lambda(x)$, $\mathcal{A}_\mu \to \mathcal{A}_\mu - \partial_\mu\Lambda$. Setting $\mathcal{A}_\mu = 0$ recovers equation 62.

2.6.2 The Potential Generated by y

The coordinate transformations of this paper supply a natural candidate,

$$\boxed{\mathcal{A}_\mu \equiv \partial_\mu y} \tag{75}$$

evaluated in the rest frame of the source, where $y = y(x)$ is static. Equation 65 then gives

$$\mathcal{A}_0 = 0, \qquad \mathcal{A}_i = \beta\gamma\,\hat{n}_i. \tag{76}$$

The quantity $\partial_\tau y = v\,\partial_x y$ used earlier is the rate of change of y along the worldline of the moving observer. It is the directional derivative $u^\mu\mathcal{A}_\mu$, not a component $\mathcal{A}_0$ of the field gradient. Three identities follow:

$$\gamma^2 = 1 + \mathcal{A}_i\mathcal{A}^i \tag{77}$$

$$\eta^{\mu\nu}\mathcal{A}_\mu\mathcal{A}_\nu = \beta^2\gamma^2 = \gamma^2 - 1, \qquad \mathcal{A}_\mu w^\mu = 0 \tag{78}$$

$$\mathcal{F}_{\mu\nu} = \partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu = 0 \tag{79}$$

where w^μ is the four-velocity of an observer at rest with respect to the source. Equation 77 is equation 26 restated, and equation 78 says the potential is a spacelike covector orthogonal to w^μ. Equation 79 holds because $\mathcal{A}_\mu$ is a pure gradient: in this sector the potential is pure gauge and carries no field strength.

2.6.3 Recovering the Induced Geometry

Transform to the frame in which the fibre coordinate is constant on a temporal surface, $y' = y - y(x)$, so that $dy' = 0$ on $\mathbf{S}$ and $\mathcal{A}_\mu = \partial_\mu y$. Equation 73 then restricts to the surface as

$$h_{\mu\nu} = g_{\mu\nu} + G_{44}\,\mathcal{A}_\mu \mathcal{A}_\nu. \tag{80}$$

For $g_{ij} = \delta_{ij}$ and $G_{44} = 1$ this gives $h_{ij} = \delta_{ij} + \partial_i y\,\partial_j y$. Along a single direction it reproduces $dS^2 = \gamma^2 dx^2$, and for a radial profile $y(R)$ it reproduces

$$A(R) = h_{RR} = 1 + \left(\frac{dy}{dR}\right)^2, \tag{81}$$

the induced radial component of the embedding section. The construction developed so far is therefore the pure-gauge sector $\mathcal{F}_{\mu\nu} = 0$ of the full five-dimensional metric.

Remark. The orbit equations in this paper are geodesics of the intrinsic metric $h_{\mu\nu}$ on $\mathbf{S}$, that is, motion constrained to the surface. They are not geodesics of the five-dimensional metric. In conventional Kaluza–Klein theory a free particle conserves $p_y = G_{44}\,(\dot{y} + \mathcal{A}_\mu \dot{x}^\mu)$, which plays the role of charge, and $\mathcal{F}_{\mu\nu}$ acts as a force. Neither appears here, and this constraint should be stated explicitly.

2.6.4 Curvature of the Surface and the Source

A hypersurface $y(x)$ of flat $\mathbb{R}^4$ has extrinsic curvature $K_{ij} = \partial_i \partial_j y / \sqrt{1 + \mathcal{A}_k \mathcal{A}^k}$, and its intrinsic scalar curvature follows from the Gauss equation,

$${}^{(3)}R = K^2 - K_{ij} K^{ij}. \tag{82}$$

For a spherically symmetric profile $y(R)$, define $\beta^2 \equiv \mathcal{A}_R^2/(1+\mathcal{A}_R^2) = 1 - \gamma^{-2}$. Then

$${}^{(3)}R = \frac{2}{R^2}\frac{d}{dR}\left(R\,\beta^2\right). \tag{83}$$

Imposing the time-symmetric Hamiltonian constraint of general relativity, ${}^{(3)}R = 16\pi G\rho/c^2$, and integrating from a regular centre gives

$$\boxed{\beta^2(R) = \frac{2G\,m(R)}{c^2 R}, \qquad m(R) = \int_0^R 4\pi\rho R'^2\,dR'} \tag{84}$$

so βc is the escape velocity from the enclosed mass. The constraint is imported from general relativity; what the embedding adds is its expression through the Hessian of y, which ties mass density to the second derivative of the fourth coordinate. Three cases follow:

- Exterior, $m = M$: $\beta^2 = r_s/R$ and $y = \pm 2\sqrt{r_s(R - r_s)} + C$, the paraboloid found above.
- Uniform density: $\beta^2 = R^2/L^2$ with $L^2 = 3c^2/(8\pi G\rho)$, so $y = \pm\sqrt{L^2 - R^2} + C$, a spherical cap of a three-sphere of radius L. This is the spatial geometry of the interior Schwarzschild solution.
- Critical density: $\rho = 3H^2/(8\pi G)$ gives $L = c/H$ and $\beta = HR/c$, the Hubble flow. The Schwarzschild radius of a critical-density ball of radius c/H equals c/H.

Gravitational binding and cosmic expansion are then the same relation $\beta^2 = 2Gm/(c^2R)$, applied to different mass distributions. The sign of the associated flow differs: infall for a bound mass, outflow for the Hubble flow.

2.6.5 The Temporal Relation

Because $\mathcal{A}_0 = 0$ in the source frame, $h_{00} = g_{00}$: the embedding alone produces no gravitational redshift. This is expected, since the Schwarzschild spacetime is a class-two manifold, with a flat embedding known only in six dimensions [21], and cannot be embedded as a hypersurface of five-dimensional flat space. The temporal function must therefore be supplied separately. In terms of the potential, the simplest closure is

$$\boxed{N(R) = \frac{1}{\gamma} = (1 + \mathcal{A}_\mu \mathcal{A}^\mu)^{-1/2}} \tag{85}$$

which is equivalent to $N^2 = 1/A(R)$ and to the statement that the time-dilation factor is the Lorentz factor of the local escape velocity. It is a postulate, not a derivation. With it,

$$\vec{g} = \frac{c^2}{2}\nabla\beta^2, \qquad \delta(M, R) = \frac{c^2}{2}\frac{\partial^2\beta^2}{\partial R^2} = \frac{2GM}{R^3}, \tag{86}$$

which reproduces $\delta(M, R)$ from the equivalence-principle section as a derivative of a dimensionless potential. The same closure gives $v^2_{\text{local}} = GM/(R - r_s)$ in the exterior region, which is $\beta^2c^2/2$ to leading order. Inside extended matter equation 85 is not the general-relativistic relation, so it should be applied to exterior regions only. The interior and time-dependent cases are treated in section 2.10.

2.6.6 Non-Gradient Component

The general potential is

$$\mathcal{A}_\mu = \partial_\mu y + a_\mu, \qquad \mathcal{F}_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu. \tag{87}$$

All results above correspond to $a_\mu = 0$. A non-zero a_μ is the only way for the five-dimensional geometry to carry a field strength, and $\mathcal{F}_{\mu\nu}$ acts on a body only through its fifth momentum p_y. We speculate that the spin dependence $\Gamma(\sigma)$ enters here as a curl component analogous to a gravitomagnetic potential, and possibly the electromagnetic parameters of section 2.2. Neither connection is derived here.

2.7 Radial Embedding and Relativistic Orbital Motion

Consider a point mass M located at the origin of a spherically symmetric spatial geometry. We assume that the radial coordinate R is associated with a corresponding position along an additional spatial coordinate y, such that the embedded hypersurface is described by

$$y = y(R; M) \tag{88}$$

Consequently, the differential displacement along the additional coordinate is

$$dy = \frac{dy}{dR}\, dR \tag{89}$$

We begin with a flat four-dimensional ambient spatial metric, restricted to the equatorial plane:

$$d\ell^2 = dR^2 + R^2 d\phi^2 + dy^2 \tag{90}$$

Substituting the radial embedding yields the induced spatial line element

$$d\ell^2_{\text{ind}} = dR^2 + R^2 d\phi^2 + \left(\frac{dy}{dR}\right)^2 dR^2 \tag{91}$$

$$= \left[1 + \left(\frac{dy}{dR}\right)^2\right] dR^2 + R^2 d\phi^2 \tag{92}$$

Defining the induced radial metric component as

$$A(R) = 1 + \left(\frac{dy}{dR}\right)^2 \tag{93}$$

the resulting metric becomes

$$\boxed{d\ell^2_{\text{ind}} = A(R) dR^2 + R^2 d\phi^2} \tag{94}$$

The radial metric component therefore contains the contribution from the gradient of the additional spatial coordinate. In particular,

$$\boxed{g^{\text{ind}}_{RR} = 1 + \left(\frac{dy}{dR}\right)^2} \tag{95}$$

2.7.1 The Orbit Equation

We introduce the reciprocal radial coordinate

$$u(\phi) = \frac{1}{R(\phi)}. \tag{96}$$

To describe geodesic motion on the induced spatial geometry, we introduce an affine parameter λ along the geodesic. The parameter λ is used solely to parameterize the

spatial trajectory and is not an additional physical coordinate or the time-equivalent quantity τ introduced previously.

A dot denotes differentiation with respect to λ:

$$\dot{R} = \frac{dR}{d\lambda}, \qquad \dot{\phi} = \frac{d\phi}{d\lambda}. \tag{97}$$

For the induced spatial metric

$$d\ell^2_{\text{ind}} = A(R)dR^2 + R^2 d\phi^2, \tag{98}$$

the geodesic Lagrangian is

$$\mathcal{L} = \frac{1}{2}\left[A(R)\dot{R}^2 + R^2\dot{\phi}^2\right]. \tag{99}$$

Since the metric does not explicitly depend on ϕ, the corresponding coordinate is cyclic. Its conjugate momentum is therefore conserved:

$$L = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = R^2\dot{\phi} = \text{constant}. \tag{100}$$

Using $u = 1/R$, this becomes

$$\dot{\phi} = Lu^2. \tag{101}$$

We now differentiate $R = 1/u$ with respect to the affine parameter λ:

$$\begin{aligned} \dot{R} &= \frac{dR}{d\phi}\dot{\phi} \\ &= -\frac{1}{u^2}\frac{du}{d\phi}Lu^2. \end{aligned} \tag{102}$$

Defining primes to denote differentiation with respect to ϕ, we obtain

$$\boxed{\dot{R} = -Lu'} \tag{103}$$

A second differentiation with respect to λ gives

$$\begin{aligned} \ddot{R} &= -L\frac{du'}{d\lambda} \\ &= -Lu''\dot{\phi} \\ &= -L^2u^2u''. \end{aligned} \tag{104}$$

Consequently,

$$\boxed{\ddot{R} = -L^2u^2u''} \tag{105}$$

Substituting these expressions into the radial geodesic equation produces

$$-L^2u^2u'' + \frac{A'}{2A}L^2(u')^2 - \frac{L^2u^3}{A} = 0. \tag{106}$$

Dividing by $-L^2u^2$ gives the orbit equation of the induced spatial geometry:

$$\boxed{u'' + \frac{u}{A(1/u)} - \frac{A'(1/u)}{2A(1/u)u^2}(u')^2 = 0} \tag{107}$$

This equation describes spatial geodesics on the embedded hypersurface. It does not, by itself, describe relativistic particle orbits in spacetime, since no temporal metric component has yet been introduced.

Substituting these expressions into the radial geodesic equation produces

$$-L^2u^2u'' + \frac{A'}{2A}L^2(u')^2 - \frac{L^2u^3}{A} = 0 \tag{108}$$

Dividing by $-L^2u^2$ gives the orbit equation of the induced spatial geometry:

$$\boxed{u'' + \frac{u}{A(1/u)} - \frac{A'(1/u)}{2A(1/u)u^2}(u')^2 = 0} \tag{109}$$

This equation describes spatial geodesics on the embedded hypersurface. It does not, by itself, describe relativistic particle orbits in spacetime, since no temporal metric component has yet been introduced.

2.7.2 The Schwarzschild Spatial Embedding

We can compare the induced radial metric with the spatial part of the Schwarzschild solution. At constant coordinate time and in the equatorial plane, the Schwarzschild spatial line element is

$$d\ell^2_{\text{Sch}} = \frac{dR^2}{1 - \dfrac{2GM}{c^2R}} + R^2d\phi^2 \tag{110}$$

Identifying the radial components of this metric with the induced radial metric gives

$$1 + \left(\frac{dy}{dR}\right)^2 = \frac{1}{1 - \dfrac{2GM}{c^2R}} \tag{111}$$

Solving for the embedding gradient,

$$\left(\frac{dy}{dR}\right)^2 = \frac{1}{1 - \dfrac{2GM}{c^2R}} - 1 \tag{112}$$

$$= \frac{\dfrac{2GM}{c^2R}}{1 - \dfrac{2GM}{c^2R}} \tag{113}$$

Introducing the Schwarzschild radius,

$$r_s = \frac{2GM}{c^2} \tag{114}$$

we obtain

$$\boxed{\frac{dy}{dR} = \pm\sqrt{\frac{r_s}{R - r_s}}} \tag{115}$$

Integration with respect to R gives

$$y(R) = \pm\sqrt{r_s} \int \frac{dR}{\sqrt{R - r_s}} \tag{116}$$

$$= \pm 2\sqrt{r_s(R - r_s)} + C \tag{117}$$

Consequently, the embedding profile is

$$\boxed{y(R) = \pm 2\sqrt{\frac{2GM}{c^2}\left(R - \frac{2GM}{c^2}\right)} + C} \tag{118}$$

This embedding reproduces the radial geometry of a Schwarzschild spatial slice for $R > r_s$. It demonstrates that the radial metric component can be represented geometrically through the gradient of an additional spatial coordinate.

However, reproducing the spatial geometry does not yet establish the complete relativistic dynamics. To investigate orbital motion, the temporal metric component must also be included.

2.7.3 The Full Spacetime Metric

Consider a static, spherically symmetric spacetime metric restricted to the equatorial plane:

$$ds^2 = -N(R)^2 c^2 dt^2 + A(R) dR^2 + R^2 d\phi^2 \tag{119}$$

Here, $A(R)$ is the radial metric component obtained from the spatial geometry, while $N(R)$ is the temporal metric function.

For the Schwarzschild solution,

$$A(R) = \frac{1}{1 - \dfrac{2GM}{c^2 R}} \tag{120}$$

and

$$N(R)^2 = 1 - \frac{2GM}{c^2 R} \tag{121}$$

The relationship between these functions is therefore

$$\boxed{N(R)^2 = \frac{1}{A(R)}} \tag{122}$$

The spatial embedding determines $A(R)$ through the radial gradient of y. The corresponding temporal relation is an additional metric condition; it does not follow from the spatial embedding alone.

2.7.4 Conserved Quantities and the Radial Equation

For a massive particle, we take the proper time τ as the affine parameter. The geodesic Lagrangian is

$$\mathcal{L} = \frac{1}{2}\left[-N(R)^2 c^2 \dot{t}^{\,2} + A(R)\dot{R}^{\,2} + R^2 \dot{\phi}^{\,2}\right] \tag{123}$$

Stationarity and spherical symmetry give two conserved quantities:

$$E = N(R)^2 c^2 \dot{t} \tag{124}$$

and

$$L = R^2 \dot{\phi} \tag{125}$$

The normalization of the four-velocity is

$$g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu = -c^2 \tag{126}$$

Substituting the metric and conserved quantities,

$$-N^2 c^2 \dot{t}^{\,2} + A\dot{R}^{\,2} + \frac{L^2}{R^2} = -c^2 \tag{127}$$

Since

$$\dot{t} = \frac{E}{N^2 c^2} \tag{128}$$

we obtain

$$-\frac{E^2}{N^2 c^2} + A\dot{R}^{\,2} + \frac{L^2}{R^2} = -c^2 \tag{129}$$

Solving for the radial velocity,

$$\boxed{A(R)\dot{R}^{\,2} = \frac{E^2}{N(R)^2 c^2} - c^2 - \frac{L^2}{R^2}} \tag{130}$$

This first integral describes the radial motion in terms of the two conserved quantities.

2.7.5 Derivation of the Relativistic Orbital Correction

We now substitute the Schwarzschild metric functions,

$$N(R)^2 = 1 - \frac{r_s}{R}, \qquad A(R) = \frac{1}{1 - \dfrac{r_s}{R}} \tag{131}$$

where $r_s = 2GM/c^2$.

The radial first integral becomes

$$\dot{R}^2 = \frac{E^2}{c^2} - \left(1 - \frac{r_s}{R}\right)\left(c^2 + \frac{L^2}{R^2}\right) \tag{132}$$

Introducing $u = 1/R$ and using

$$\dot{R} = -Lu', \tag{133}$$

we obtain

$$L^2(u')^2 = \frac{E^2}{c^2} - (1 - r_s u)(c^2 + L^2u^2) \tag{134}$$

Expanding the right-hand side,

$$L^2(u')^2 = \frac{E^2}{c^2} - c^2 - L^2u^2 + r_s c^2 u + r_s L^2 u^3 \tag{135}$$

Differentiating with respect to ϕ gives

$$2L^2u'u'' = -2L^2uu' + r_s c^2 u' + 3r_s L^2 u^2 u' \tag{136}$$

For a nonconstant radial trajectory, dividing by $2L^2u'$ yields

$$u'' = -u + \frac{r_s c^2}{2L^2} + \frac{3r_s}{2}u^2 \tag{137}$$

Rearranging,

$$u'' + u = \frac{r_s c^2}{2L^2} + \frac{3r_s}{2}u^2 \tag{138}$$

Finally, substituting $r_s = 2GM/c^2$, we arrive at

$$\boxed{\frac{d^2u}{d\phi^2} + u = \frac{GM}{L^2} + \frac{3GM}{c^2}u^2} \tag{139}$$

The first term on the right-hand side is the Newtonian contribution. The second term is the leading relativistic orbital correction.

In the Newtonian limit, $c \to \infty$, the relativistic contribution vanishes and the equation reduces to

$$\frac{d^2u}{d\phi^2} + u = \frac{GM}{L^2} \tag{140}$$

Its solution describes a conic-section orbit:

$$u(\phi) = \frac{GM}{L^2}\left[1 + e\cos(\phi - \phi_0)\right] \tag{141}$$

where e is the orbital eccentricity and ϕ_0 specifies the orientation of the orbit.

2.8 Circular Orbital Velocity and Stability

We now derive the circular orbital velocity directly from the spacetime metric. This provides an independent relationship between the metric components and the motion of a test particle, while also allowing us to establish the conditions under which circular orbits are stable.

We begin with the general static, spherically symmetric metric, restricted to the equatorial plane:

$$ds^2 = -N(R)^2c^2dt^2 + A(R)dR^2 + R^2d\phi^2 \tag{142}$$

Here, $N(R)$ represents the temporal metric function and $A(R)$ represents the radial metric component. Both are functions of the radial coordinate R.

2.8.1 Geodesic Equations

We parameterize the trajectory of a massive test particle by its proper time τ. The geodesic Lagrangian is

$$\mathcal{L} = \frac{1}{2}\left[-N(R)^2c^2\dot{t}^{\,2} + A(R)\dot{R}^{\,2} + R^2\dot{\phi}^{\,2}\right] \tag{143}$$

where a dot denotes differentiation with respect to τ.

Because the metric is independent of both t and ϕ, these coordinates are cyclic. Their corresponding conjugate momenta are conserved. We define the conserved energy per unit mass and specific angular momentum as

$$E = N(R)^2c^2\dot{t}, \tag{144}$$

$$L = R^2\dot{\phi}. \tag{145}$$

Solving for the coordinate derivatives gives

$$\dot{t} = \frac{E}{N(R)^2c^2}, \tag{146}$$

$$\dot{\phi} = \frac{L}{R^2}. \tag{147}$$

We now apply the Euler–Lagrange equation to the radial coordinate:

$$\frac{d}{d\tau}\left(\frac{\partial\mathcal{L}}{\partial\dot{R}}\right) - \frac{\partial\mathcal{L}}{\partial R} = 0 \tag{148}$$

Evaluating the derivatives gives

$$\frac{d}{d\tau}\left(A(R)\dot{R}\right) + N(R)N'(R)c^2\dot{t}^{\,2} - \frac{1}{2}A'(R)\dot{R}^{\,2} - R\dot{\phi}^{\,2} = 0 \tag{149}$$

Expanding the total derivative and dividing by $A(R)$ yields the radial geodesic equation:

$$\ddot{R} + \frac{A'(R)}{2A(R)}\dot{R}^{\,2} + \frac{N(R)N'(R)c^2}{A(R)}\dot{t}^{\,2} - \frac{R}{A(R)}\dot{\phi}^{\,2} = 0 \tag{150}$$

2.8.2 Circular Orbital Velocity

For a circular orbit, the radial coordinate is constant. Therefore,

$$\dot{R} = 0, \qquad \ddot{R} = 0 \tag{151}$$

Substituting these conditions into the radial geodesic equation gives

$$\frac{N(R)N'(R)c^2}{A(R)}\dot{t}^{\,2} - \frac{R}{A(R)}\dot{\phi}^{\,2} = 0 \tag{152}$$

Multiplying by $A(R)$ and rearranging,

$$R\dot{\phi}^{\,2} = N(R)N'(R)c^2\dot{t}^{\,2} \tag{153}$$

The coordinate angular velocity is

$$\Omega = \frac{d\phi}{dt} = \frac{\dot{\phi}}{\dot{t}} \tag{154}$$

Dividing by $\dot{t}^{\,2}$, we obtain

$$\boxed{\Omega^2 = \frac{c^2 N(R)N'(R)}{R}} \tag{155}$$

To obtain the velocity measured by a local static observer, we must account for the relationship between coordinate time and proper time. At fixed radial coordinate, the metric becomes

$$ds^2 = -N(R)^2c^2dt^2 \tag{156}$$

Consequently, the observer's proper time satisfies

$$d\tau_{\text{static}} = N(R)dt \tag{157}$$

The local circumferential distance traversed by the particle is

$$d\ell = R\,d\phi \tag{158}$$

The locally measured orbital velocity is therefore

$$v_{\text{local}} = \frac{d\ell}{d\tau_{\text{static}}} = \frac{R}{N(R)}\frac{d\phi}{dt} \tag{159}$$

Squaring and substituting the expression for Ω^2 gives

$$v_{\text{local}}^2 = \frac{R^2}{N(R)^2}\Omega^2 \tag{160}$$

$$= \frac{R^2}{N(R)^2}\frac{c^2 N(R)N'(R)}{R}. \tag{161}$$

Thus, the locally measured circular orbital velocity is

$$\boxed{v_{\text{local}}^2 = c^2 R\frac{N'(R)}{N(R)}} \tag{162}$$

This result expresses circular orbital velocity directly in terms of the radial derivative of the temporal metric function.

2.8.3 Effective Potential and Orbital Stability

We next examine the conditions under which a circular orbit is stable. For a massive particle, the four-velocity normalization is

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -c^2 \tag{163}$$

Substituting the metric components gives

$$-N(R)^2c^2\dot{t}^{\,2} + A(R)\dot{R}^{\,2} + R^2\dot{\phi}^{\,2} = -c^2 \tag{164}$$

Using the conserved quantities,

$$E = N(R)^2c^2\dot{t}, \tag{165}$$

$$L = R^2\dot{\phi}, \tag{166}$$

we obtain

$$-\frac{E^2}{N(R)^2c^2} + A(R)\dot{R}^{\,2} + \frac{L^2}{R^2} = -c^2 \tag{167}$$

Multiplying by $N(R)^2$ and rearranging,

$$N(R)^2A(R)\dot{R}^{\,2} + N(R)^2\left(c^2 + \frac{L^2}{R^2}\right) = \frac{E^2}{c^2} \tag{168}$$

We identify the effective potential as

$$\boxed{V_{\text{eff}}(R) = N(R)^2\left(c^2 + \frac{L^2}{R^2}\right)} \tag{169}$$

The radial equation can now be written as

$$N(R)^2A(R)\dot{R}^{\,2} + V_{\text{eff}}(R) = \frac{E^2}{c^2} \tag{170}$$

A circular orbit occurs at an extremum of the effective potential:

$$\frac{dV_{\text{eff}}}{dR} = 0 \tag{171}$$

Differentiating while holding L constant gives

$$\frac{dV_{\text{eff}}}{dR} = 2N(R)N'(R)\left(c^2 + \frac{L^2}{R^2}\right) - \frac{2N(R)^2L^2}{R^3} \tag{172}$$

Setting this expression equal to zero and solving for L^2 yields

$$\boxed{L^2 = \frac{c^2R^3\left(N(R)^2\right)'}{2N(R)^2 - R\left(N(R)^2\right)'}} \tag{173}$$

A circular orbit is stable when the effective potential has a local minimum. The stability condition is therefore

$$\boxed{\frac{d^2V_{\text{eff}}}{dR^2} > 0} \tag{174}$$

If the second derivative is negative, the circular orbit is unstable. If it vanishes, the orbit is marginally stable and higher-order terms must be examined.

2.8.4 Schwarzschild Reference Case

We can verify these expressions using the Schwarzschild temporal metric function:

$$N(R)^2 = 1 - \frac{2GM}{c^2R} \tag{175}$$

Defining the Schwarzschild radius,

$$r_s = \frac{2GM}{c^2} \tag{176}$$

we can write

$$N(R)^2 = 1 - \frac{r_s}{R} \tag{177}$$

Differentiating,

$$\left(N(R)^2\right)' = \frac{r_s}{R^2} \tag{178}$$

Since

$$\frac{N'(R)}{N(R)} = \frac{\left(N(R)^2\right)'}{2N(R)^2} \tag{179}$$

the locally measured circular velocity becomes

$$v_{\text{local}}^2 = \frac{c^2R}{2}\frac{\left(N(R)^2\right)'}{N(R)^2} \tag{180}$$

Substituting the Schwarzschild function,

$$v_{\text{local}}^2 = \frac{c^2 R}{2} \frac{r_s/R^2}{1 - r_s/R} \tag{181}$$

$$= \frac{c^2 r_s}{2(R - r_s)}. \tag{182}$$

Therefore,

$$\boxed{v_{\text{local}}^2 = \frac{GM}{R - \frac{2GM}{c^2}}} \tag{183}$$

In the weak-field limit,

$$\frac{2GM}{c^2 R} \ll 1 \tag{184}$$

this expression reduces to the Newtonian circular orbital velocity:

$$v_{\text{local}} \approx \sqrt{\frac{GM}{R}} \tag{185}$$

We can also evaluate the angular momentum required for a Schwarzschild circular orbit. Substitution into the general expression gives

$$L^2 = \frac{c^2 R^3 (r_s/R^2)}{2(1 - r_s/R) - r_s/R} \tag{186}$$

$$= \frac{c^2 r_s R^2}{2R - 3r_s}. \tag{187}$$

Equivalently,

$$\boxed{L^2 = \frac{GMR^2}{R - \frac{3GM}{c^2}}} \tag{188}$$

Evaluating the second derivative of the effective potential at the circular-orbit condition gives

$$\boxed{V''_{\text{eff}}(R) = \frac{2c^2 r_s (R - 3r_s)}{R^3 (2R - 3r_s)}} \tag{189}$$

The existence of a timelike circular orbit requires

$$R > \frac{3}{2} r_s = \frac{3GM}{c^2} \tag{190}$$

The additional stability condition requires

$$R > 3r_s = \frac{6GM}{c^2} \tag{191}$$

Thus, the innermost stable circular orbit in Schwarzschild geometry is

$$\boxed{R_{\mathrm{ISCO}} = \frac{6GM}{c^2}} \tag{192}$$

2.9 Implications for the Higher-Dimensional Metric

The derivation establishes an important distinction between the radial and temporal components of the metric. The radial component determines the radial contribution to the line element, while the circular orbital velocity depends on the temporal metric function:

$$v_{\mathrm{local}}^2 = c^2 R \frac{N'(R)}{N(R)} \tag{193}$$

In the spatial embedding developed in this work, the radial metric component is related to the additional spatial coordinate through

$$A(R) = 1 + \left(\frac{dy}{dR}\right)^2 \tag{194}$$

This relationship determines the radial geometry of the embedded spatial surface. It does not independently determine the temporal metric function $N(R)$.

A complete derivation of circular orbital motion within the proposed higher-dimensional geometry therefore requires establishing $N(R)$ from the higher-dimensional construction. In particular, the relationship

$$N(R)^2 = \frac{1}{A(R)} \tag{195}$$

must be derived independently if it is to be used; it cannot be inferred solely from the radial embedding.

Once both metric components have been established, the general expressions derived above can be used to determine the predicted orbital velocity, angular momentum, and stability conditions.

2.10 Relation to the Painlevé–Gullstrand Form and to Cosmology

The identification $\beta^2 = 2Gm(R)/(c^2R)$ of equation 84 is not new in itself. This section places it in its established context, gives a dictionary between the quantities of this paper and their standard counterparts, and states what the construction adds and what it does not.

2.10.1 Schwarzschild in Painlevé–Gullstrand form

With $\beta^2 = r_s/R$ the Schwarzschild metric can be written [22–24]

$$ds^2 = -c^2 dT^2 + (dR + \beta c\, dT)^2 + R^2 d\Omega^2, \tag{196}$$

where T is the proper time of observers in radial free fall from rest at infinity. The surfaces $T = \text{const}$ are intrinsically flat, and βc is the Newtonian escape velocity, so space can be pictured as a flat background flowing inward at βc [25]. The rest space of an observer held at fixed R has radial metric component

$$h_{RR} = g_{RR} - \frac{g_{TR}^2}{g_{TT}} = 1 + \beta^2\gamma^2 = \gamma^2, \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}}, \tag{197}$$

so γ is the Lorentz factor between the infalling frame and the static observers, and the static proper length is $d\ell = \gamma\, dR$. This is the content of equations $dS^2 = dx^2 + dy^2 = \gamma^2 dx^2$ and $dy = \beta\gamma\, dx$ above, with x the coordinate R of the flat slices, y the Flamm embedding [26] of the static slice, and $\partial_R y = \beta\gamma$.

Along a free-fall worldline $dR/dT = -\beta c$, so $dy/dT = -c\beta^2\gamma$. With $v = \beta c$ its magnitude is $v^2/\sqrt{c^2 - v^2}$, which is equation 60, and its sign agrees with $c\, dt' = -\partial_\tau y$. The time-equivalent parameter τ therefore plays the role of the proper time T of the flow.

Temporal relation. For any $\beta(R)$, the change of time $dT = dt + \beta\, dR/\left[c(1-\beta^2)\right]$ takes equation 196 to

$$ds^2 = -(1-\beta^2)c^2 dt^2 + \frac{dR^2}{1-\beta^2} + R^2 d\Omega^2, \tag{198}$$

so $N^2 = 1/A = 1 - \beta^2$. The postulate of equation 85 is therefore equivalent to requiring that the frame moving at βc relative to the static observers has unit lapse and flat slices. For pressureless matter this is exact, because the dust frame is geodesic.

2.10.2 Hubble flow

For flat FLRW, $R = a(t)r$ gives $a\, dr = dR - HR\, dt$ and

$$ds^2 = -c^2 dt^2 + (dR - \beta_H c\, dt)^2 + R^2 d\Omega^2, \qquad \beta_H = \frac{HR}{c}, \tag{199}$$

the same form as equation 196 with the opposite sign of the shift. With $m = \frac{4\pi}{3}\rho R^3$ and the Friedmann equation $H^2 = 8\pi G\rho/3$, $\beta_H^2 = 2Gm/(c^2 R)$. The Hubble radius $R = c/H$ is where $\beta = 1$ and $2Gm/c^2 = R$, which is the apparent horizon of the flat FLRW universe [27, 28]. The Newtonian counterpart is the zero-energy condition of Milne and McCrea [29, 30]: the Hubble velocity equals the escape velocity from the enclosed mass.

2.10.3 The general relation

For spherical, pressureless, marginally bound dust (Lemaître–Tolman–Bondi [31–33] with zero energy function), $R'\, dr = dR - \dot{R}\, dt$ with $\dot{R}^2 = 2Gm/R$ gives equation

196 with $\beta^2 = 2Gm(R,T)/(c^2R)$ and m the Misner–Sharp mass [34]. The generalised Painlevé–Gullstrand form extends this to pressure [35]. Equation 84 is the same relation. Schwarzschild and flat FLRW are two cases of it:

Table 2 Cases of $\beta^2 = 2Gm/(c^2R)$ with unit lapse and flat slices

Configuration	$m(R)$	Flow	Static-slice embedding
Exterior of a mass	M	inward	paraboloid
Uniform density ball	$\frac{4\pi}{3}\rho R^3$	inward or outward	cap of a 3-sphere, $L^2 = 3c^2/8\pi G\rho$
Critical density (flat FLRW)	$\frac{4\pi}{3}\rho_c R^3$	outward	cap with $L = c/H$

Binding and expansion are then one relation with different mass distributions, and the sign of the flow distinguishes infall from Hubble expansion. The statement that gravitational acceleration and cosmic inflation are the same process, as observed from different frames, is made precise in this sense.

2.10.4 Dictionary

Table 3 Quantities of this paper and their standard counterparts

This paper	Standard counterpart
$\beta c = v$	shift vector ("river velocity"), escape velocity from the enclosed mass
γ	Lorentz factor between the flow and static observers; $1/\gamma$ is the static lapse
$y(R)$, $\partial_R y = \beta\gamma$	Flamm embedding of the static slice, $d\ell^2 = dR^2 + dy^2$
$\partial_\tau y = v^2/\sqrt{c^2 - v^2}$	dy/dT along a free-fall worldline
$\delta(M,R) = 2GM/R^3$	radial tidal eigenvalue
v_{orbit}	βc, the escape velocity of the enclosed mass
v_{cosmo}	$c\,(v_{\text{esc}}/R)\,\tau$, with v_{esc}/R the Hubble-like rate of the flow at R
$v_{\text{kinematic}}$	$c\,\Omega\,\tau$ with $\Omega = \sqrt{GM/R^3}$ the circular-orbit angular frequency

The last two rows follow from $v = c\sqrt{1-\varphi^{-2}} \approx c\sqrt{2(\varphi-1)}$ with $\varphi - 1 = GM/R^3$ and $\frac{1}{2}GM/R^3$ respectively, and they explain why the two velocities differ by a factor $\sqrt{2}$, the ratio of escape to circular speed.

Remark. The same factor applies to equation 49. The velocity obtained there is βc, the escape velocity of the enclosed mass. In Schwarzschild geometry the circular velocity is $\beta c/\sqrt{2}$ to leading order. This is the origin of the discrepancy noted earlier between $v \approx v_{\text{escape}}$ and a stable orbit, and a derivation of the inertial correction is needed before equation 49 can be read as a circular velocity.

2.10.5 Scope

In the regimes above (vacuum or pressureless matter, spherical symmetry, flat-slice flow) the construction is equivalent to general relativity. The relation $\beta^2 = 2Gm/(c^2R)$ is taken from its Hamiltonian constraint, and $\dot{R}^2 = 2Gm/R$ is the Newtonian energy equation, so neither is derived here. What the construction adds is a reading of the flow as a coordinate gradient of a fourth spatial coordinate, with the Lorentz factor of the flow as the slope, together with the identification of the time-like parameter with the proper time of the flow. Predictions that differ from general relativity must come from sectors outside these regimes: a non-gradient potential a_μ, a varying G_{44}, or the relations of the equivalence-principle section.

2.10.6 Perihelion Precession

To determine the effect of the relativistic term, define the semilatus rectum

$$p = \frac{L^2}{GM} \tag{200}$$

The relativistic orbit equation becomes

$$u'' + u = \frac{1}{p} + \frac{3GM}{c^2}u^2 \tag{201}$$

In the weak-field limit,

$$\frac{GM}{pc^2} \ll 1 \tag{202}$$

we seek a first-order approximation to the Newtonian orbit. Substituting the leading-order form into the relativistic contribution gives the approximate solution

$$u(\phi) \simeq \frac{1}{p}\left[1 + e\cos\left(\left(1 - \frac{3GM}{pc^2}\right)\phi\right)\right] \tag{203}$$

The radial oscillation therefore does not repeat after an angular displacement of exactly 2π. Instead, the perihelion advances by

$$\Delta\phi = 2\pi\left[\frac{1}{1 - \dfrac{3GM}{pc^2}} - 1\right] \tag{204}$$

To leading order,

$$\boxed{\Delta\phi = \frac{6\pi GM}{pc^2} = \frac{6\pi GM}{a(1-e^2)c^2}} \tag{205}$$

where a is the orbital semimajor axis and e is the eccentricity.
This is the leading-order Schwarzschild perihelion advance.

2.10.7 Geometric Interpretation

The derivation establishes a connection between the radial embedding and relativistic orbital geometry. The additional spatial coordinate produces an induced radial metric component,

$$A(R) = 1 + \left(\frac{dy}{dR}\right)^2, \tag{206}$$

which can reproduce the radial geometry of a Schwarzschild spatial slice. When the corresponding temporal metric function is also specified, the resulting spacetime geodesics produce the relativistic orbital correction.

The distinction between these two contributions is essential. The spatial embedding determines the radial geometry, whereas the temporal metric determines how the geometry relates to the evolution of a massive particle. The relativistic orbital correction follows from their combined effect, rather than from the spatial embedding alone.

Consequently, reproducing the Schwarzschild orbit equation through this construction requires establishing both the radial embedding and the corresponding temporal metric relation. Whether that temporal relation can be derived independently from the proposed fifth-dimensional geometry remains a separate question.

3 Discussion

3.1 The Issue of Dimensionality

Several places throughout this paper we make calculations that appear to be dependent upon our choice of units, particularly that of time. We propose that this is purely an artifact of the math; left over residuals from a derivative that was never an inherent part of the system. Instead we interpret this geometry purely as magnitudes of length, and time as the ratio of lengths.

When we keep the reference length fixed where $l = \widehat{x} = |\vec{x}|$, we find that

$$\frac{dx}{dt} \equiv \frac{dx}{d\tau} = \frac{dx}{\left(\frac{\dot{x}}{\widehat{x}}\right)} = \widehat{x} \tag{207}$$

3.2 The Appearance of Electromagnetic Parameters

We do not claim to derive a direct mechanical derivation for the fine structure constant, nor an explanation for it's appearance in section 2.2. We simply apply the coordinate transformations defined in section 2 and discover that α appears repeatedly. While the probability is difficult to quantize, we feel that this repeated appearance of α, appearing in gravitational approximations for all 3 of the local bodies with boundary like characteristics likely indicates a shared function which itself contains or produces α.

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